TEAM EXECUTION BLUEPRINT • TARGET: INNOHACKS 4.0 & BUILD WITH BHARAT

CascadeGuard AI: Multi-Surface Engineering Blueprint

Autonomous Lateral Social Engineering & Account Takeover (ATO) Interceptor built "At the Glass" across WhatsApp Web, Telegram, and Android.

📌 1. Strategic Context: Why "Mitigation 4" (Multi-Surface)?

When attackers compromise accounts via QR phishing, SIM swapping, or rogue web sessions, **they weaponize established trust**. Traditional antivirus is blind because messages originate from authentic phone numbers; messaging platforms are blind because End-to-End Encryption (E2EE) prevents server inspection.

To eliminate platform ToS ban risks and provide defense across all user contexts, **CascadeGuard implements a Multi-Surface Strategy**:

SURFACE 1 WhatsApp Web (DOM)

Mechanism: Chromium Extension (Manifest v3) using a passive `MutationObserver` on stable accessibility nodes (`[role="row"]`).
ToS Defense: Zero clicks, zero WebSocket hooks. In-line shield injected into an isolated Shadow DOM overlay. Works like Grammarly.

SURFACE 2 Telegram Ecosystem

Mechanism: Official Open-Source TDLib / MTProto Client Plugin and Telegram Bot Security Gateway.
ToS Defense: Telegram's client is 100% open-source and officially supports custom clients without scraping or policy bans.

SURFACE 3 Android OS Companion

Mechanism: Android `NotificationListenerService` & `AccessibilityService`.
ToS Defense: Operates at the OS notification/intent layer (standard Truecaller / Google Play Protect security model). Zero app modification.

⚙️ 2. Multi-Agent Engine Architecture

The core engine performs a parallel dual-engine audit in **< 300 milliseconds** entirely on-device:

Agent 1: Stylometric Profiler

- Cosine distance against local non-reversible user baseline
- Hinglish vernacular ratio & colloquial slang tracking
- Punctuation entropy (exclamations, comma shifts)
- Casing pressure velocity (all-caps urgency bursts)

Agent 2: Forensic & Intent Scanner

- Financial coercion & pressure keywords heuristics
- UPI VPA regex extractor & QR payload parsing
- Simulated I4C / NPCI Mule Account Registry query
- Malicious APK & phishing contest links scanner

Agent 3: Autonomous Challenger & Interlock

- Deploys in-line warning banner in Shadow DOM
- Locks target UPI links & renders buttons unclickable
- Generates Zero-Knowledge "Friend Quiz" verification challenge

Agent 4: Blast-Radius Containment

- Maps real-time viral propagation attack topology
- Flags compromised accounts in college & batch groups
- Issues containment signal to group admins (zero chat text)

3. Detection & Enforcement in Action

CascadeGuard operates directly at the glass. When an attacker sends a coercive payment or phishing message from a compromised account, the defense shield deploys instantaneously:



Figure 1: CascadeGuard UI — In-Line Stylometric Drift Warning, Payment Interlock & Security HUD

INTERLOCK **UPI Payment Lock**

Automatically intercepts clicks on UPI payment links or QR codes, disabling direct transfers to unverified mule handles.

CHALLENGE **Zero-Knowledge Quiz**

Prompts target with a private relational challenge ("What was our hostel warden's nickname?") that an imposter cannot answer.

TELEMETRY **Security HUD**

Displays real-time vector gauges for Stylometric Drift, Financial Coercion, Urgency Score, and Edge Processing Latency (<300ms).

4. Network Contagion & Community Defense

When a compromised student account attacks downstream peers and college WhatsApp groups, CascadeGuard isolates the infection topology:

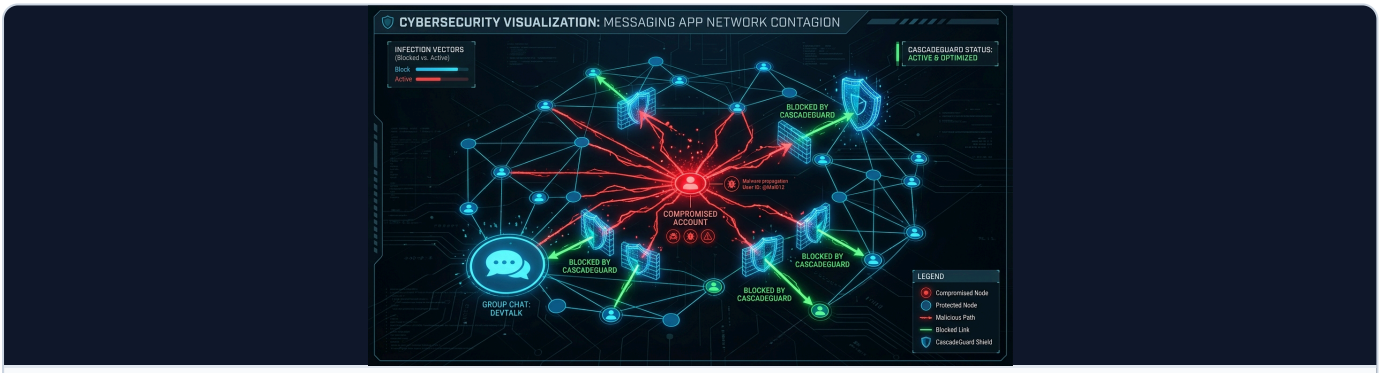


Figure 2: Network Contagion Graph — Isolating Hijacked Nodes and Halting Lateral Blast Radius

> **Blast-Radius Isolation in Action:** When Aman's account attempts to send a malicious ₹5,000 scam to Priya and a voting APK to Rahul, both nodes are shielded. The CS Batch WhatsApp Group admin receives an automated quarantine alert, preventing lateral infection to 64 students.

5. 24-Hour Hackathon Execution Timeline

Phase & Hours	Milestone / Objectives	Key Engineering Deliverables	Primary Lead
Phase 1 `Hours 0 - 6`	Core Intelligence Engine & Heuristics Calibration	• Mathematical stylometric drift vector calculations. • Hinglish slang tokenizer & vocabulary cosine distance. • Regex VPA extractor & simulated NPCI mule threat feed.	**Member 2** AI / NLP Lead
Phase 2 `Hours 6 - 12`	Surface 1: Web Extension & Live Simulator	• Dual-screen chat UI (Attacker console vs. Protected Victim). • In-line Shadow DOM interception shield. • Locked UPI payment barrier & test scenario presets.	**Member 1** Frontend Lead
Phase 3 `Hours 12 - 18`	Surface 2 & 3: Multi-Surface Integration	• Telegram open-source TDLib plugin / bot interceptor demo. • Android NotificationListenerService companion stub. • Autonomous "Friend Quiz" verification challenge modal.	**Member 3** Full-Stack Lead
Phase 4 `Hours 18 - 21`	Contagion Graph & DPDP Act Audit	• Dynamic topological SVG contagion network graph. • Group admin quarantine telemetry dispatcher. • DPDP Act 2023 Statutory Compliance audit console.	**Member 3 & 4** Systems & Product
Phase 5 `Hours 21 - 24`	Pitch Rehearsal, Deck & Q&A Readiness	• Finalize 10-slide Round 1 PPT submission deck. • Live 3-minute pitch choreography & demo backup video. • Rehearse tough judge questions (anti-ban, privacy, false positives).	**Member 4** Product & Pitch Lead

6. Team Role Allocation Matrix (4 Members)

`MEMBER 1` **Frontend & UX Lead**

Responsibilities: Dual-screen chat simulator, WhatsApp Web presentation styling, Shadow DOM overlay injection, and preset attack scenarios.
Key Output: Crisp, responsive web simulator ready for live judge evaluation.

`MEMBER 2` **AI / NLP & Stylometry Lead**

Responsibilities: Mathematical stylometric drift pipeline, Hinglish/English dialect ratios, coercion heuristics scoring, and VPA threat feeds.
Key Output: Sub-50ms deterministic edge inference engine.

`MEMBER 3` **Full-Stack & Multi-Surface Lead**

Responsibilities: Network contagion blast-radius visualizer, Telegram TDLib integration, Android companion stub, and NPCI API connector.
Key Output: Contagion network map and multi-surface proof of concept.

`MEMBER 4` **Product, Pitch & Compliance Lead**

Responsibilities: Round 1 presentation deck, DPDP Act 2023 legal compliance framework, pitch script choreography, and Q&A defense playbook.
Key Output: Winning pitch deck and smooth presentation flow.

7. Privacy & DPDP Act 2023 Compliance Moat

Statutory Principle	DPDP Act 2023 Requirement	CascadeGuard AI Implementation
Data Minimization	Collect only strictly necessary data	Zero persistent chat storage. Ephemeral RAM disposal after <300ms audit.
Purpose Limitation	Process data strictly for stated security purpose	Processing restricted purely to client-side lateral ATO anomaly scoring.
No Cloud Egress	Prevent unauthorized personal data transfer	0 Bytes of chat text leave the device. Manifest v3 `connect-src 'none'`.
Irreversible Vectors	Prevent recreation of personal communications	Baselines stored as non-reversible mathematical statistical vectors.

🎤 8. The Winning 3-Minute Hackathon Pitch Script

Minute 1: The Hook & The Problem (Weaponized Trust)

"Respected judges, how many of you have received an urgent WhatsApp message from a friend saying: 'Bhai emergency hai, hospital me hu, please transfer ₹5,000'? In India, account hijacking isn't about data theft—it is about weaponizing established interpersonal trust to execute cascading scams. Traditional antivirus is blind because messages come from authentic phone numbers, and WhatsApp servers cannot inspect chats due to End-to-End Encryption."

Minute 2: The Live Demonstration (300ms Interception)

"Watch what happens when an attacker sends a medical scam from our friend Rohit's compromised account. In under 300 milliseconds, CascadeGuard's on-device engine detects a 91% stylometric drift, extracts the suspicious UPI VPA, engages our Payment Interlock to disable direct payment, and generates a zero-knowledge 'Friend Quiz' verification challenge. Across the college hostel network, the attack is instantly quarantined before money is lost."

Minute 3: Privacy, Bharat Scale & The Moat

"Best of all, zero chat text ever leaves the user's device. Our Multi-Surface Architecture protects users on WhatsApp Web, Telegram, and Android OS without violating platform terms of service. By storing non-reversible mathematical fingerprints and enforcing zero network egress, CascadeGuard is 100% compliant with India's DPDP Act 2023. We are building the immune system for Bharat's digital communication."

🎯 9. Tough Judge Q&A Defense Playbook

Q: "Won't WhatsApp ban users for injecting into their web application?"

Answer: Meta bans tools that automate messaging, scrape contacts, or hook internal WebSockets. CascadeGuard operates strictly at the presentation layer—the exact same way Grammarly, 1Password, or screen-reader accessibility tools do. We use a read-only `MutationObserver` on stable accessibility nodes and render our shield inside an isolated Shadow DOM without touching WhatsApp's React tree.

Q: "Isn't a browser extension with chat access a massive spyware risk?"

Answer: Under Manifest v3, our Content Security Policy strictly enforces `connect-src 'none'`. It is cryptographically impossible for the extension to transmit messages to external servers. All text processing occurs in volatile RAM and is immediately discarded. Zero bytes of chat text leave the device.

Q: "What happens if a friend actually has a genuine emergency?"

Answer: CascadeGuard never blocks communications or deletes messages. It simply engages a payment interlock and offers a private relational challenge ("What was our hostel warden's nickname?"). A real friend answers in seconds, instantly unlocking the payment channel.